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Efficient Data Management in SAS®: Using Macro Statements and DO Loops to Combine and Merge Datasets

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Abstract
Monitoring vital signs such as body temperature and heart rate is critical for managing pre-term infants. Continuous abdominal (central) and foot (peripheral) skin temperature measurements enable calculation of a thermal gradient, a potential early marker of infection. Heart rate characteristics (HRC) scores also predict infection but require a specialized HeRO monitor. Our multicenter study compared thermal gradients and HRC scores in infants born before 32 weeks' gestation. A total of 364 infants were enrolled across five sites Duke (n=25), ECU (n=137), MUSC (n=31), Prisma (n=149), and UNC (n=11) under NIH/NINR grant 1R01NR017872, with IRB approval and parental consent. Each infant had thermistor probes placed on the abdomen and foot to continuously record skin temperature. Thermistor probes recorded continuous abdominal and foot skin temperatures, while temperature and HRC data were collected via a research datalogger and the HeRO monitor, then converted into Excel. Each infant's dataset averaged approximately 41,000 rows. Individual SAS datasets were created and merged using macro and do-loop programs in SAS 9.4, incorporating demographics and consolidating data across sites for analysis.

Keywords: SAS, body temperature, infant

Background
Monitoring physiological vital signs such as body temperature and heart rate is critical for guiding treatment and optimizing health outcomes in preterm infants. One approach to assessing thermal regulation is the continuous measurement of both central (abdominal) and peripheral (foot) skin temperatures, which provides insight into blood perfusion across the infant's body.[1] Abnormal thermal gradients—defined as a central-peripheral temperature difference (CPtd) greater than 2°C or a negative CPtd (where foot temperature exceeds abdominal temperature)—have been associated with the onset of infection and physiological stress in premature infants.[2]

Another vital sign used to detect early signs of infection is the heart rate characteristics (HRC) score, derived from heart rate variability and provided by the HeRO monitor (Predictive Medical Science Corp., Charlottesville, VA).[3] This score is calculated from the infant's standard cardiopulmonary monitoring and has been shown to be predictive of sepsis risk.

This study, funded by the NIH/NINR (Grant 1R01NR017872),[4] builds upon Dail's ongoing research on the use of thermal gradients as indicators of autonomic nervous system dysregulation and early predictors of infection in very preterm infants. HRC scores were used as a comparator to evaluate the predictive value of thermal gradients over the first 28 days of each infant's life.

Purpose
The purpose of this paper is to demonstrate how SAS macros and DO loops can be used to combine and merge datasets from multiple sites in a temperature study.

Methodology
A total of 364 preterm infants—born at ≤32 weeks gestational age and weighing less than 1500 grams at birth were enrolled in the study following Institutional Review Board approval and parental consent. Physiological temperature data were continuously recorded during the first 28 days of life using research-grade dataloggers and later downloaded via USB.

Each infant had two skin temperature probes (thermistors): one placed on the abdomen (central temperature) and one on the sole of the foot (peripheral temperature). Standard clinical care involved maintaining body temperature between

36.5°C and 37.0°C using a servo-controlled heated incubator, regulated by a separate clinical abdominal thermistor. For research purposes, temperature data were collected every minute over the 28-day period. In parallel, heart rate characteristic (HRC) scores were recorded hourly using the HeRO monitor, which was connected to each infant's cardiopulmonary monitoring system.

Research nurses were responsible for downloading both temperature and HRC data, which were then transferred to a data manager. The data manager merged the files into a single Excel file per infant, including the following variables: date, time, minutes since birth (MSB), abdominal temperature (ABD), foot temperature (FT), date and time of temperature measurement (DLD, DT), HeRO monitor date and time (DH, TH), and HRC score.

In addition, demographic information was extracted from each infant's electronic medical record and entered a REDCap database. Data were collected across five study sites: Duke (n=25), East Carolina University (ECU, n=137), Medical University of South Carolina (MUSC, n=31), Prisma Health (n=149), and the University of North Carolina (UNC, n=11). Each infant's dataset contained approximately 41,002 observations.

SAS datasets were created for each infant, and a SAS macro was used to automate the merging of all infant data within each site. Demographic variables were then merged with the site-level datasets, and finally, data from all sites were consolidated into a single master dataset In this paper, we present the syntax used for the Duke site only. A macro statement, along with a DO loop, was written to streamline the process of combining and merging these datasets using SAS/STAT® statistical software, version 9.4.[3]

Data Steps

Table 1 provides an example of a portion of data for a single infant in an Excel file. We use the duke site to demonstrate the process of combining and merging data for all infants.

Table1 Example of data in Excel

id	msb	Date	Time	dld	dt	abd	ftp	dh	Th	HRC
421_002	253	9/30/2019	10:54:00	9/30/2019	10:54:00	36.5	35.67	.	.	.
421_002	254	9/30/2019	10:55:00	9/30/2019	10:55:00	36.53	35.7	.	.	.
421_002	255	9/30/2019	10:56:00	9/30/2019	10:56:00	36.54	35.7	.	.	.
421_002	256	9/30/2019	10:57:00	9/30/2019	10:57:00	36.54	35.67	.	.	.
421_002	257	9/30/2019	10:58:00	9/30/2019	10:58:00	36.53	35.66	.	.	.
421_002	258	9/30/2019	10:59:00	9/30/2019	10:59:00	36.52	35.64	.	.	.
421_002	259	9/30/2019	11:00:00	9/30/2019	11:00:00	36.53	35.67	.	.	.
421_002	260	9/30/2019	11:01:00	9/30/2019	11:01:00	36.53	35.65	.	.	.
421_002	261	9/30/2019	11:02:00	9/30/2019	11:02:00	36.52	35.65	.	.	.
421_002	262	9/30/2019	11:03:00	9/30/2019	11:03:00	36.51	35.64	.	.	.
421_002	263	9/30/2019	11:04:00	9/30/2019	11:04:00	36.48	35.62	.	.	.
421_002	264	9/30/2019	11:05:00	9/30/2019	11:05:00	36.36	35.58	.	.	.
421_002	265	9/30/2019	11:06:00	9/30/2019	11:06:00	36.4	35.57	.	.	.
421_002	266	9/30/2019	11:07:00	9/30/2019	11:07:00	36.45	35.55	.	.	.
421_002	267	9/30/2019	11:08:00	9/30/2019	11:08:00	36.41	35.54	.	.	.
421_002	268	9/30/2019	11:09:00	9/30/2019	11:09:00	36.41	35.53	.	.	.
421_002	269	9/30/2019	11:10:00	9/30/2019	11:10:00	36.41	35.52	.	.	.
421_002	270	9/30/2019	11:11:00	9/30/2019	11:11:00	36.42	35.51	.	.	.
421_002	271	9/30/2019	11:12:00	9/30/2019	11:12:00	36.42	35.5	.	.	.
421_002	272	9/30/2019	11:13:00	9/30/2019	11:13:00	36.42	35.49	.	.	.
421_002	273	9/30/2019	11:14:00	9/30/2019	11:14:00	36.43	35.48	.	.	.

421_002	**274**	**9/30/2019**	**11:15:00**	**9/30/2019**	**11:15:00**	**36.43**	**35.46**	**.**	**.**	**.**
421_002	**275**	**9/30/2019**	**11:16:00**	**9/30/2019**	**11:16:00**	**36.43**	**35.45**	**.**	**.**	**.**

Table 2 presents the SAS macro program (without a DO loop) used to generate unique ID numbers and annotate data. Each ID is composed of a three-digit site code followed by a three-digit infant identifier.

Table 2. Combine and create id numbers for all infants for Duke site without making a statement

SAS Program

```
libname r24 'd:\r24\data\duke\sasdata\';
%macro data (d=d2 , s=r24.duke2 , i=421002);
data &d;
set &s;
iid=&i;
run;
%mend data;run;
%data (d=d2 , s=r24.duke2 , i=421002);
%data (d=d3 , s=r24.duke3 , i=421003);
%data (d=d4 , s=r24.duke4 , i=421004);
%data (d=d5 , s=r24.duke5 , i=421005);
%data (d=d6 , s=r24.duke6 , i=421006);
%data (d=d7 , s=r24.duke7 , i=421007);
%data (d=d8 , s=r24.duke8 , i=421008);
%data (d=d9 , s=r24.duke9 , i=421009);
%data (d=d10, s=r24.duke10, i=421010);
%data (d=d11, s=r24.duke11, i=421011);
%data (d=d12, s=r24.duke12, i=421012);
%data (d=d13, s=r24.duke13, i=421013);
%data (d=d14, s=r24.duke14, i=421014);
%data (d=d15, s=r24.duke15, i=421015);
%data (d=d16, s=r24.duke16, i=421016);
%data (d=d17, s=r24.duke17, i=421017);
%data (d=d18, s=r24.duke18, i=421018);
%data (d=d19, s=r24.duke19, i=421019);
%data (d=d20, s=r24.duke20, i=421020);
%data (d=d21, s=r24.duke21, i=421021);
%data (d=d22, s=r24.duke22, i=421022);
%data (d=d23, s=r24.duke23, i=421023);
%data (d=d24, s=r24.duke24, i=421024);
%data (d=d25, s=r24.duke25, i=421025);
%data (d=d26, s=r24.duke26, i=421026);run;
```

Current code calls the %data macro repeatedly with a patterned structure: only the numbers in the dataset names (d2 to d26, r24.duke2 to r24.duke26, and the iid values from 421002 to 421026) change incrementally.

Table 3 displays the SAS macro program, without a DO loop, used to generate unique ID numbers and annotate data. Each ID combines a three-digit site code with a three-digit infant identifier.

SAS Program

```
libname r24 'd:\r24\data\duke\sasdata\';

%macro data (d=d2 , s=r24.duke2 , i=421002);
data &d;
set &s;
iid=&i;
run;
%mend data;
```

```
run;
%macro gendata(start=2, end=26);
    %do i = &start %to &end;
        data d&i;
            set r24.duke&i;
            iid = %eval(421000 + &i);
        run;
    %end;
%mend;
%gendata(start=2, end=26);
run;
```

%macro generate_data(start=2, end=26); – A macro to generate datasets from d2 to d26.

%do i = &start %to &end; – Loop through the specified range.

data d&i; – Dynamically names the output dataset.

set r24.duke&i; – Dynamically sets the input dataset.

iid = %eval(421000 + &i); – Calculates the iid value corresponding to the current loop index.

%generate_data(...) – Calls the macro to execute the loop.

Table 4 presents the program that merges individual infant datasets into a single consolidated dataset. This program is used to generate and combine data for infants across all sites.

Table 4. Combine and create id numbers for all infants for Duke site without making a statement

SAS Program

```
data final;
set d2-d26;
by id;
if first.id then idn=1;
 else idn+1;
run;
libname r24 'd:\r24\data\duke\sasdata\';
data r24.dukeall
(keep= id iid DATE TIME msb dld dt abd ftp dh th hrc) ;
retain id iid DATE TIME msb dld dt abd ftp dh th hrc;
 set final;
 format abd ftp hrc 8.2; run;
```

Table 5 shows the SAS program used to merge the combined Duke site data with demographic information. The same program is applied to merge combined data and demographics for infants from all other sites.

Table 5. SAS Program to merge combined files with demographic file.

SAS Program

```
libname r24 'd:\r24\data\duke\sasdata\';

data one;
 set r24.dukeall;
 site=1;
proc sort data=one;by id; run;
```

```
libname r24 'd:\r24\data\demo\';


data two;
 set r24.mdemo;
proc sort data=two;
by id;
run;

data all;
merge one (in=a) two (in=b);
by id;run;

data r24.dukedemo;
set all;
run;
```

Combined libname paths to the common root folder d:\r24\data\ to avoid repeating.
Removed unnecessary extra data and set steps.
Combined output dataset directly into the target library folder r24.combine.
Kept sorting steps separate since merge requires sorted datasets.

Table 6 shows the SAS program to merge all sites method 1.

Table 6 SAS program to merge all sites method 1.

SAS Program

```
proc format;
value sitef 1="Duke"  2="ECU" 3="MUSC" 4="Prisma" 5="UNC"                ;
libname duke 'd:\r24\data\duke\combine\';
libname ecu 'd:\r24\data\ecu\combine\';
libname musc 'd:\r24\data\musc\combine\';
libname prisma 'd:\r24\data\prisma\combine\';
libname unc 'd:\r24\data\unc\combine\';

data all;
  set duke.dukedemo   ecu.ecudemo    musc.muscdemo   prisma.prismademo
      unc.uncdemo;
  format site sitef.;run;

libname r24 'd:\r24\sasdata\';

data r24.r24all;
  set all;
run;
```

Table 7 SAS program to merge all sites method 2.

SAS Program

```
%macro combine;
  %local sites i;
  %let sites=duke ecu musc prisma unc;
  /* Assign libnames */
  %do i=1 %to %sysfunc(countw(&sites));
    %let site=%scan(&sites, &i);
```

```
    libname &site "d:\r24\data\&site.\combine\";
  %end;

  data all;
    set
    %do i=1 %to %sysfunc(countw(&sites));
      %let site=%scan(&sites, &i);
      &site..&site.demo
    %end;
    ;
    format site sitef.;
  run;

  libname r24 'd:\r24\sasdata\';

  data r24.r24all;
    set all;
  run;
%mend;

%combine;
run;
```

The macro %combine_sites automates the process of:

1. Assigning libnames to the folders for each study site.
2. Combining datasets from all those sites into one large dataset.
3. Saving the combined dataset to a specified library.

%macro combine_sites;This starts the macro definition called combine_sites.
%local sites i; Declares two local macro variables: sites (to hold the list of site names) and i (a counter for looping). %let sites=duke ecu musc prisma unc; Defines the list of study sites you want to process.

```
/* Assign libnames */
  %do i=1 %to %sysfunc(countw(&sites));
    %let site=%scan(&sites, &i);
    libname &site "d:\r24\data\&site.\combine\";
  %end;
```

This is a loop that runs from 1 to the number of sites.
%sysfunc(countw(&sites)) counts how many site names are in sites. %scan(&sites, &i) extracts the i-th site name.
libname &site ... dynamically assigns a libname for each site folder (e.g., libname duke "d:\r24\data\duke\combine\";).

```
data all;
    set
    %do i=1 %to %sysfunc(countw(&sites));
      %let site=%scan(&sites, &i);
      &site..&site.demo
    %end;
    ;
    format site sitef.;
  run;
```

Creates a new dataset called all. Uses a loop inside the set statement to read all datasets for each site (e.g., duke.dukedemo, ecu.ecudemo, etc.). Combines them vertically (one after the other). Applies the format sitef. to the site variable.

```
libname r24 'd:\r24\sasdata\';
  data r24.r24all;
    set all;  run;
```

Assigns a libname to the destination folder. Saves the combined dataset all into r24.r24all.

%mend;
Ends the macro definition.
%combine_sites;
Calls (runs) the macro.

Table 8 displays the SAS program used to generate descriptive statistics. The PROC FREQ and PROC MEANS procedures are applied to produce descriptive statistics stratified by site and ID.

Table8 SAS Procedure to run descriptive statistics.

SAS Program

```
Ods rtf;
ods listing close;
proc sort data =one; by  id;
%macro freqb (d,s,i,r,c,t);
proc freq data=&d;
where site=&s;
       tables  (&i)*(&r)*(&c)  ;
     title " frequency table/&t";
       %mend freqb;run;
%freqb (one,1,id,day, cptdg group, duke by id day);
%freqb (one,1,id,week, cptdg group, duke by id week);
%freqb (one,2,id,day, cptdg group, ECU by id day);
%freqb (one,2,id,week, cptdg group, ECU by id week);
%freqb (one,3,id,day, cptdg group, MUSC by id day);
%freqb (one,3,id,week, cptdg group, MUSC by id week);
%freqb (one,4,id,day, cptdg group, Prisma by id day);
%freqb (one,4,id,week, cptdg group, Prisma by id week);
%freqb (one,5,id,day, cptdg group, UNC by id day);
%freqb (one,5,id,week, cptdg group, UNC by id week);
by  id;
run;

Ods rtf;
ods listing close;
%macro avg (d,q,v,t);

proc means data=&d n mean std min max  maxdec=2;
  class &q;
     var  &v  ;
     title " means /by  &t" ;run;

%mend avg;
%avg(one, site,cptd abd ftp hrc gagew gaged bwt, by site);
 %avg(one, id ,cptd abd ftp hrc gagew gaged bwt, by id);
%avg(one, site id ,cptd abd ftp hrc gagew gaged bwt, by site id);
%avg(one, site id day ,cptd abd ftp hrc gagew gaged bwt, by site id day);
%avg(one, site id week ,cptd abd ftp hrc gagew gaged bwt, by site id week );
ods rtf close;ods listing;quit;run;
```

Conclusion

Data collected for the Infant Temperature study underwent several processing steps to prepare for analysis. SAS datasets were created for each infant, with unique ID numbers assigned. A macro was then used to combine data from all infants

within each site. Demographic variables were merged with the combined site data, and finally, datasets from all sites were consolidated.

Using macros in SAS offers key advantages of automation: Eliminates repetitive coding by looping through sites automatically, flexibility: Easily add or remove sites by updating a list, and maintainability: Centralized code is easier to update and debug.

Future improvements should focus on enhancing computing efficiency by incorporating DO loops within macros, reducing the number of macro calls and running times. Currently, infant IDs are assigned manually, which can be tedious and prone to error as the dataset grows. Implementing loops to assign IDs dynamically would streamline this process. Additionally, applying conditional logic to assign site IDs would improve code readability and maintainability. Finally, optimizing the macro in Table 5 by using a variable list of sites would allow it to process multiple sites in one call instead of running separately for each site, further improving efficiency.

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